

WEEK:- 10

Topic: Fast API Framework, REST APIs, Backend Development, and API Testing

Objectives:

To understand the Fast API framework, develop RESTful APIs, learn backend development concepts, and perform API testing to build reliable and scalable web applications.

Learning Outcomes:

- Learned the fundamentals of the FastAPI framework and backend development.
- Understood REST API architecture and endpoint creation.
- Gained knowledge of request handling, response management, and data validation.
- Developed skills in API testing and debugging techniques.
- Improved backend development and system integration capabilities.

WORK SUMMARY:

During Week 10, I worked on the FastAPI framework, REST APIs, backend development, and API testing. The training focused on understanding how backend systems process requests, manage data, and communicate with frontend applications. I attended technical sessions and participated in hands-on exercises to learn the architecture and working principles of modern web APIs. Through practical implementation activities, I explored how FastAPI simplifies backend development by providing high performance, automatic documentation, and efficient request handling capabilities.

Throughout the week, I gained practical experience in creating API endpoints, handling HTTP requests and responses, and implementing business logic within backend applications. I learned how to define routes, process user inputs, validate data, and return structured responses in JSON format. Special emphasis was placed on understanding RESTful API design principles and how APIs facilitate communication between different software modules. I also explored database interaction concepts and the role of backend services in supporting application functionality.

In addition, I performed API testing using various testing tools and techniques to verify endpoint functionality, validate responses, and identify potential issues. Debugging exercises and troubleshooting sessions helped me improve my analytical thinking and problem-solving abilities. Regular reviews and mentor feedback provided valuable insights into industry-standard backend development practices.

The knowledge and practical experience gained during this week significantly strengthened my understanding of web services and server-side application development. It provided a solid foundation for building scalable applications, integrating systems, and developing complete full-stack solutions in future projects.

Activities Performed:

- Developed REST API endpoints using the FastAPI framework.
- Performed API testing, debugging, and backend functionality validation